Data Visualization Projects and Learning

Cooper Union STEM Saturdays · Inventor's Lab: Data Visualization · 2024

Final project Comparing U.S. employment patterns

What the project presents

This project organizes U.S. employment data into charts that let an audience compare changes across age groups, education levels, and gender. The completed work includes an Excel file with seven worksheets, a 22-slide presentation, and bilingual speaking notes. Overall trends, selected close-ups, and difference charts let viewers follow the comparisons without reading the original table row by row.

From the table to the presentation

The workbook retains 44 annual rows labelled 1979–2022 and 122 fields. Separate worksheets organize age, education, gender, and related subgroup comparisons. The age section compares people aged 16–24, 25–54, 55–64, and 65+. The education section compares five categories and calculates the advanced-degree minus bachelor's-degree gap. The gender section compares men and women, including differences within age groups. Line charts show changes over time, bar charts compare groups, and close-ups support discussion of selected periods.

An example of the comparison

In the saved education table, the advanced-degree group is 82.8% in 1979 and the bachelor's-degree group is 75.0%, a difference of 7.8 percentage points. The saved difference for 2015 is 0.3 points. Plotting the yearly difference separately makes the distance between the two groups easier to inspect than comparing two lines alone. This describes the table's pattern; it does not establish that educational expansion or a particular policy caused the change.

Presentation and collaboration

The talk progresses through age, education, and gender before summarizing the observations. Bilingual notes retain speaker assignments, vocabulary annotations, and rehearsal cues. The original deck credits Shaoxuan Xiong as producer and Richard Feng as editor. Year alignment and some percentage summaries in the archived work still need review; the repository records the specific issues in its data notes.

GitHub repository

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What I practiced during the course

Organizing data and calculating meaningful summaries

I practiced distinguishing numeric fields from categories, organizing observations by year or group, and using Excel to calculate counts, averages, standard deviations, and frequencies. Passenger records, coin-mass measurements, and annual employment ratios require different summaries. The exercises gave me concrete practice distinguishing an individual observation, a group percentage, and a total across years.

Choosing a chart for the question

Line charts helped me compare several employment groups over time. Frequency charts in the coin and Titanic exercises showed the distribution of measurements, ages, or fares. Bar charts supported comparisons of groups and gaps. The settlement-time exercise represented the same process-stage data with different charts, giving me a way to compare how clearly each chart showed the overall total and the relative size of its components.

Connecting the calculation to the comparison

I practiced moving from raw values to differences and proportions: subtracting the bachelor's-degree ratio from the advanced-degree ratio, or comparing men and women within the same year. The archive also gives a concrete example of why the denominator matters. The share of survivors who travelled first class is different from the survival rate among first-class passengers. These exercises provide a basis for checking years, units, denominators, and aggregation when revisiting an analysis.

Explaining the analysis to an audience

The final project connected spreadsheet work with oral explanation. I practiced introducing a question, showing a chart, identifying an observation, and explaining how that observation related to the question. Bilingual rehearsal notes and a collaborative presentation also involved dividing speaking sections, connecting the parts of the talk, and describing groups, time ranges, and changes in English. The saved work documents that practice while leaving historical interpretations open to further checking.

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How the classroom exercises support the final project

Titanic From passenger records to summaries

Using 891 passenger records, this exercise summarizes survival counts, gender composition, age and fare distributions, and the passenger-class composition of survivors. Counts, averages, standard deviation, frequency calculations, and a PivotTable provide practice in organizing records into comparisons.

Penny mass From measurements to distributions

The workbook uses coin year, mass, and mint labels to calculate averages, standard deviation, frequencies, and normal-density values. Separate worksheets retain measurements and subsets. The exercise connects individual values with a distribution and shows why an average alone does not fully describe a dataset.

Settlement time Comparing ways to show the same data

Two charts show total settlement time and four process components: notification, documentation or inspection, review, and waiting. The table supplies actual rows for 2015–2022 and a forecast row for 2023. The exercise concerns visual representation; the saved workbook does not contain a forecasting model.

Program selection From comparison to recommendation

A five-slide presentation compares program popularity, audience groups, and production costs to explain which programs to retain. This connects visual interpretation with a recommendation. The underlying audience/cost workbook was not preserved with the materials, so that limitation is recorded rather than filling in missing calculations.

GitHub repository